every possible way of getting there. This is known as Feynman's "sum-over-paths" approach to quantum mechanics.7

At this point your classical upbringing is balking: How can one electron simultaneously take different paths—and no less than an infinite number of them? This seems like a defensible objection, but quantum mechanics—the physics of our world—requires that you hold such pedestrian complaints in abeyance. The result of calculations using Feynman's approach agree with those of the wave function method, which agree with experiments. You must allow nature to dictate what is and what is not sensible. As Feynman once wrote, "[Quantum mechanics] describes nature as absurd from the point of view of common sense. And it fully agrees with experiment. So I hope you can accept nature as She is—absurd."8

But no matter how absurd nature is when examined on microscopic scales, things must conspire so that we recover the familiar prosaic happenings of the world experienced on everyday scales. To this end, Feynman showed that if you examine the motion of large objects—like baseballs, airplanes, or planets, all large in comparison with subatomic particles—his rule for assigning numbers to each path ensures that all paths but one cancel each other out when their contributions are combined. In effect, only one of the infinity of paths matters as far as the motion of the object is concerned. And this trajectory is precisely the one emerging from Newton's laws of motion. This is why in the everyday world it seems to us that objects—like a ball tossed in the air—follow a single, unique, and predictable trajectory from their origin to their destination. But for microscopic objects, Feynman's rule for assigning numbers to paths shows that many different paths can and often do contribute to an object's motion. In the double-slit experiment, for example, some of these paths pass through different slits, giving rise to the interference pattern observed. In the microscopic realm we therefore cannot assert that an electron passes through only one slit or the other. The interference pattern and Feynman's alternative formulation of quantum mechanics emphatically attest to the contrary.

Just as we may find that varying interpretations of a book or a film can be more or less helpful in aiding our understanding of different aspects of the work, the same is true of the different approaches to